VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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MINI PROJECT

on

" SignBridge: A Real-Time Bidirectional Sign Language Communication System with a Unique Hand Gesture Dataset "

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I. Title of the Mini Project

" SignBridge: A Real-Time Bidirectional Sign Language Communication System with a Unique Hand Gesture Dataset "

II. Abstract

Communication between sign language users and non-signers remains difficult in everyday settings such as hospitals, banks, colleges, and shops, since most existing tools depend on controlled backgrounds, good lighting, or user-specific calibration to recognize signs reliably. This project proposes SignBridge, a real-time hand-gesture recognition system built on computer vision and deep learning. A standard camera captures a live video feed of the signer's hand; a background-removal step isolates the hand from the surrounding scene during preprocessing, allowing gestures to be recognized reliably even under reduced lighting conditions. A CNN extracts spatial features from the isolated hand region while an LSTM based sequence model captures temporal motion, allowing the system to recognize gestures directly from the live feed, without requiring images to be uploaded, and to work across different users despite variation in hand size, orientation, and appearance. Recognized gestures are converted into text and, optionally, synthesized speech, making the system directly useful as an accessibility tool. Unlike prior systems that rely on isolated signs, controlled backgrounds, or user-specific calibration, SignBridge unifies real-time, user-independent, hand-gesture-only recognition into a single camera-only application, aiming to make communication in healthcare, banking, education, and retail settings more accessible without dedicated sensors or a human interpreter.

III. Objectives

1. To collect a dataset of hand gestures that is not available in existing gesture datasets.
2. To train a machine learning model using the collected dataset so that it can accurately recognize static hand gestures in real time.
3. To validate the model to ensure that it provides accurate and reliable gesture recognition across different signers and lighting conditions

IV. Scope

The scope of SignBridge covers the collection of a hand gesture dataset and the training of a machine learning-based model for static gesture recognition. The system processes individual image frames captured from a standard RGB webcam or smartphone camera in real time, recognizing discrete hand gestures without specialized sensors, gloves, or markers. It is intended for service-oriented environments such as hospitals, banks, colleges, and retail counters, where sensor-free gesture recognition can support communication between sign language users and non-signers. The current scope includes dataset collection across varied users and lighting conditions, background-removal preprocessing, feature extraction, frame-level gesture classification, and conversion of recognized gestures into text and optional speech output. The prototype will be evaluated for recognition accuracy, latency, and usability under normal and low-light conditions. Continuous sentence-level signing, video-based temporal modeling, bidirectional translation, and multi-dialect support remain outside the current scope and are addressed under Future Scope.

V. Societal Impact & Outcomes of the Project

This project aims to reduce the communication difficulties faced by deaf and hard-of-hearing people in everyday situations such as healthcare, banking, education, and shopping. When

people do not understand sign language, it can lead to delays, dependence on interpreters, or difficulty communicating independently. Our system provides an affordable, camera-based, two-way communication solution that converts sign language into speech and speech into sign language in real time. This can help sign language users communicate more independently and make public spaces more accessible and inclusive. The project will result in a working real-time prototype, along with accuracy and response-time measurements and a unique gesture dataset. In the future, the system can be expanded with more signs and languages and can be used in mobile apps, public-service kiosks, and sign language learning tools.

VI. SDG Mapping:

SDG 10 – Reduced Inequalities:

Our project reduces communication barriers between sign-language users and non-sign-language users, promoting equal participation and independence.

SDG 4 Quality Education:

The system can support inclusive classrooms by enabling easier communication between students and teachers.

SDG 8 –Decent Work and Economic Growth:

It can improve communication in workplaces, helping create more inclusive employment environments.

VII. References

[1] A. O. Hashi, S. Z. M. Hashim, and A. B. Asamah, "A systematic review of hand gesture recognition: An update from 2018 to 2024," IEEE Access, vol. 12, pp. 143603–143623, 2024.

[2] U. Rastogi, R. P. Mahapatra, and S. Kumar, "Advancements in Machine Learning Techniques for Hand Gesture-Based Sign Language Recognition: A Comprehensive Review," -Archives of Computational Methods in Engineering, vol. 32, pp. 4265–4302, 2025.